WHEN DOES CELL CONDITIONING HARM RECONSTRUCTION? DIAGNOSTICS FOR ARBITRARY-SCALE SUPER-RESOLUTION

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ABSTRACT

Target-cell conditioning in arbitrary-scale super-resolution can improve average reconstruction while harming particular images or frequency bands. We investigate this discrepancy using saved natural-image outputs and controlled probes of frozen models. Low-resolution Nyquist normalisation reduces cross-scale recovery variation in all 27 model–dataset–seed groups, but pooled consistency hides image-wise differences. An exact squared-error decomposition separates harmful corrections into overshoot and wrong direction: their frequencies differ between LTE and SC-INR and between band and full-RGB evaluation. Halving the correction rescues 11/13 and 6/27 harmful band outcomes, respectively, yet increases overall RGB MSE by 2.41% and 0.97%. At fixed scale, controlled high-frequency input paths can reverse correction alignment; changing target phase at fixed power also reverses readout gain. These finite experiments show why frequency ordering and global attenuation are insufficient descriptions of readout benefit. They provide conditional failure diagnostics, rather than a new decoder, a deployable selector or a universal causal mechanism.

***Index Terms*—** Arbitrary-scale super-resolution, cell conditioning, frequency recovery, readout failure analysis

1. INTRODUCTION AND RELATED WORK

Arbitrary-scale super-resolution (ASISR) predicts image values at continuous coordinates. LIIF supplies target cell size to a local decoder [1]; LTE introduces cell-dependent phase estimation [2]. This information specifies the requested scale but does not guarantee that the resulting output change reduces reconstruction error. Our question is: *at the same scale and frequency band, why can a cell-conditioned correction help one image and hurt another?* We examine output-level failure types and controlled conditions that change benefit, without identifying an internal neural mechanism.

Harmful cell effects are not new. LIIF already varies cell size at fixed queries; IPE-LIIF integrates positional features and reports cases where an additional cell input hurts [3]. Thera supplies analytic Gaussian filtering and discusses its mismatch with bicubic training [4]. Our fixed-checkpoint interventions are not a performance comparison against these methods. Integrating intermediate features also need not equal integrating a nonlinear decoder's RGB output.

Frequency-resolved analysis has precedents in HyRA's linear/nonlinear responses [5], texture-based error analysis [6], sinusoidal degradation probes [7], and Nyquist-normalised forecasting [8]. The importance of phase beyond spectral magnitude is classical [9]. We do not claim these principles, or the quadratic identity below, as new. A wider continuous-SR benchmark studies architecture and retraining effects [10]; we isolate interventions on inherited models.

Our contribution is a connected empirical diagnosis. First, we establish a useful frequency coordinate while showing how pooling conceals image-wise variation. Second, we distinguish overshoot from wrong-direction corrections, quantify their different band/RGB consequences, and test high-frequency input paths and phase at fixed scale. These controls locate conditions missing from coarse spectral descriptions. Failed observation-only forecasts remain evidence of a gap between explanation and decision, and are evaluated separately from the descriptive findings.

2. EVALUATION OBJECTS AND FREQUENCY RECOVERY

2.1. Keep the comparison fixed

Let $y = Ax$ denote the LR observation, $Q_h x$ the desired target measurement, and c the cell supplied to $F_\theta(y, q, c)$. Changes to A, Q_h and c answer different questions. Natural-image tests use discrete HR targets and bicubic antialiased downsampling, not certified box averages of a known continuous field. Assigning $c = h$ does not establish that the model implements the physical operator Q_h.

The LIIF/LTE/SC-INR variants inherit an EDSR-baseline encoder, DIV2K training [11], 1000 epochs and training scales $\times 1$–$\times 4$. SC-INR is the examined sinc-modulated Fourier-feature implementation. A trained full/no-readout ablation changes weights; a native/zero-cell comparison holds weights, observation and queries fixed. Zero cell may be outside training support. Frozen probes use one seed1 checkpoint per LTE/SC model; this LTE snapshot omits the original paper's extrapolation cell clamp. Claims concern these implementations.

2.2. A useful coordinate with image-wise exceptions

For image i, band b, target energy $S_{ib} = \|\Pi_b(x_i - \bar{x}_i)\|^2$ and error energy $E_{ib} = \|\Pi_b(x_i - \hat{x}_i)\|^2$, define

$$R_b = 1 - \frac{\sum_i E_{ib}}{\sum_i S_{ib}}, \qquad D = \mathrm{median}_b[\max_s R_b(s) - \min_s R_b(s)]. \tag{1}$$

The frequency study uses the RGB mean, unwindowed FFTs, centre HR240 crops and respectively 14, 30 and 30 images from Set14, BSD100 and Urban100 [12, 13, 14]. With Chebyshev frequency $\nu = \max(|\nu_x|, |\nu_y|)$ and scale $s = 2, 3, 4$, seven bins span $.25, .5, \ldots, 2$ in $u = 2s\nu$; $u = 1$ is LR Nyquist. The absolute control fixes bin edges at $\nu = u/6$. Different bin supports compare evaluation coordinates, not equality of reconstructed functions.

Normalisation reduces D in all nine primary model–dataset groups, from .31–.46 to .013–.061, and all 27 groups including two additional seeds. The original requirement of $D \leq .05$ in eight of nine primary groups fails (6/9). In 24 saved-output sensitivity settings covering frequency subdomains, scale pairs and pooling, all 27 groups retain the reduction; these are dependent comparisons. On the original domain, the tolerance passes 19/27 pooled versus 27/27 equal-image groups.

To separate weighting from image-wise changes, write $R_{sb} = \sum_i w_{isb} R_{isb}$ with $w_{isb} = S_{isb}/\sum_j S_{jsb}$. Holding weights at $\bar{w}_{ib} = \frac{1}{3}\sum_s w_{isb}$ reduces D in all nine Set14 groups (.0493–.0616 to .0227–.0294) but increases it in all nine Urban100 groups (.0147–.0202 to .0293–.0538). All nine Urban100 pooled curves pass .05, while only 14–17 of 30 individual images per group pass that same descriptive tolerance. Weights can amplify or conceal variation; HR-dependent reweighting is neither a deployable rule nor an independently identified physical cause.

3. FROM FREQUENCY ORDERING TO FAILURE TYPES

3.1. Coarse ordering is an insufficient explanation

In LTE/SC seeds 2 and 3, trained full/no-readout recovery losses meet the frozen band-order criterion ($\rho \geq .5$); high-band loss exceeds middle-band loss in 34/36 dataset–scale settings. Yet the footprint score $(.00006, .00552, .07235)$ has exactly the same ranks as band index $(1, 2, 3)$. These paired tests use RGB-mean HR256 crops and three bands cut at $1/8, 1/4, 1/2$. A separate HR240 test averages full-model outputs over 31 conditioning scales $1, 1.1, \ldots, 4$ at fixed features and queries. Its ordering passes the separate ordering criterion ($\rho \geq .7$) in 4/4 groups, but passes its .8 function-similarity criterion in 0/4 and exceeds the matched footprint correlation by .05 in only 2/4. Similarity is the fraction of cases where the trained ablation is closer to the averaged than to the full output. Different crops and pooling are evaluated within their own protocols. A ranking therefore does not establish the tested integral interpretation. Nor are training ablations interchangeable with fixed-weight interventions: on barbara at $\times 2$, SC's high-band recovery gain is $-.193296$ for the trained pair but $+.009733$ for native/zero-cell.

Table 1. Saved-output failure diagnosis on 32 development images at two scales. Counts are image–scale(–band) records, not independent images. RGB MSE change is the ratio of mean errors over all 64 cases, relative to native output; harm shares weight negative gains by their magnitude.

Diagnostic	LTE	SC-INR
Harmful bands / 192	13	27
Overshoot / wrong direction	12 / 1	14 / 13
Direction share of band harm	8.91%	74.57%
Harmful full RGB / 64	1	9
Overshoot / wrong direction	1 / 0	7 / 2
Overshoot share of RGB harm	100%	99.52%
Bands rescued at $\lambda = .5$	11/13	6/27
RGB MSE change at $\lambda = .5$	+2.405%	+.966%

3.2. Overshoot and wrong direction

For fixed-weight outputs, let F_c be native-cell prediction, F_0 zero-cell prediction, t the HR target, and $d = F_c - F_0$. Inner products are pixel/channel means, optionally after band projection. Set $a = \langle t - F_0, d\rangle$, $b = \|d\|^2$ and

$$G(\lambda) = \mathrm{MSE}(F_0, t) - \mathrm{MSE}(F_0 + \lambda d, t) = 2\lambda a - \lambda^2 b. \tag{2}$$

For a harmful native correction $G(1) < 0$ and $b > 0$, $0 < a < b/2$ is *overshoot*: a smaller positive correction can help. When $a < 0$, the *direction is wrong*: every positive multiple hurts. The oracle optimum is $\lambda^* = a/b$. This classical identity diagnoses saved outputs using HR truth; output interpolation is not a cell intervention or an observation-only method. We define harm by $G(1) < -10^{-8}$. For the saved natural outputs, b, $|a|$ and $|a - b/2|$ within $64\epsilon_{64}\max(\mathrm{MSE}(F_0, t), b, 10^{-8})$ are retained as numerical boundaries; ϵ_{64} is machine precision.

We reuse all 32 old calibration images (DIV2K train0001–0032), centre HR128, $\times 2/4$, shave4 and raw RGB, with no additional inference. Each model has 64 image–scale cases and 192 non-DC band records, with Chebyshev cutoffs $1/8, 1/4, 1/2$. Band energies use the full pixel-domain normalisation, not division by the number of retained frequencies. These images are from the training split, not a new generalisation test. Table 1 separates the negative subsets in band and full-RGB evaluation.

Loss shares sum $-G(1)$ only over the corresponding negative subset. SC's wrong-direction cases dominate *band* harm, but overshoot accounts for 99.52% of *full-RGB* harm. Of cases

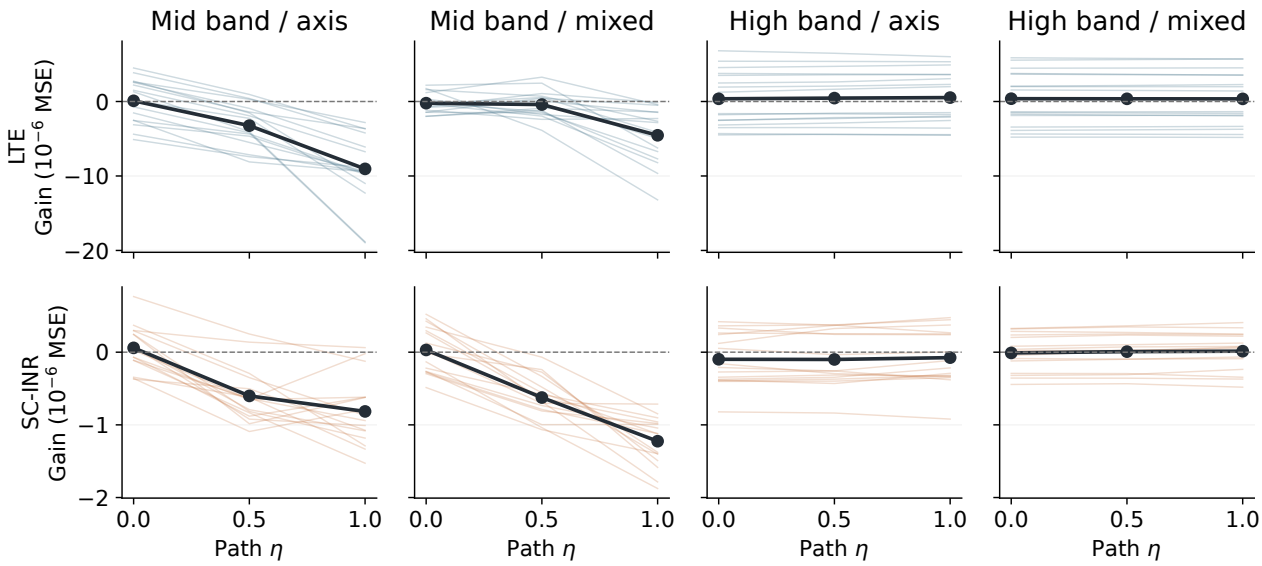


Fig. 1. All phase/position conditions of the four supra-Nyquist templates (primary centre32 region). Thin lines show 16 related phase–position conditions per panel; dark lines are finite-grid means, not confidence intervals. Columns 1–2 measure middle-band gain, columns 3–4 high-band gain, in 10^{-6} MSE units. Negative values mean native cell is worse than zero cell. These relative readout gains are not total reconstruction quality changes under the input-path intervention.

containing a negative band, 9/10 LTE and 12/21 SC cases still benefit in full RGB. Bands exclude DC, whose residual contribution is retained separately. Thus band failure cannot be treated as image failure.

Here rescue means $G(.5) > 10^{-8}$, not merely a numerical tie. Uniformly halving d rescues many harmful bands but increases overall RGB MSE (Table 1): it also weakens useful corrections. Retaining .25 or .75 likewise worsens overall MSE in both models. These fixed attenuation tests do not prove that every global weight fails; they show why a local rescue alone cannot justify a global policy.

4. CONTROLLED INPUT-PATH AND PHASE EXPERIMENTS

4.1. Design: separate two interventions

To probe conditions behind alignment changes, we keep the two checkpoints and $\times 4$ scale fixed. A monochrome HR64 target, replicated across RGB, is $t = .5 + L + H$, with three low cosine modes $L : \{(2,0),(0,2),(2,2)\}$ with zero initial phases. All six cosine modes have amplitude .04. Frequencies are cycles per 64 pixels. Four H templates are

$$M_a = \{(10,0),(12,0),(14,0)\},$$
$$M_c = \{(10,0),(0,12),(10,10)\},$$
$$H_a = \{(18,0),(22,0),(26,0)\},$$
$$H_c = \{(18,0),(0,22),(18,18)\}.$$

All lie above LR Nyquist (8 cycles). A low-frequency sham uses $\{(4,0),(0,4),(4,4)\}$. Four high-branch phase vectors (one zero and three uniform vectors, seed 20261012) are crossed with four translations $(0,0),(.5,0),(0,.5),(.5,.5)$ in LR-pixel units. Translations move both branches and are nested position controls, not independent mechanisms.

For each fixed target, a periodic separable bicubic-AA operator A produces LR16 observations

$$y_\eta = A(.5 + L) + \eta A(H), \qquad \eta \in \{0, .5, 1\}. \tag{3}$$

The sham always uses At. Thus η controls a specified high-frequency input path; it also changes LR energy/gradients, not aliasing independently of input statistics. Separately, changing high-branch phase holds each Fourier magnitude fixed while changing the target. All modes are even, so both full64 and primary centre32 regions contain integer periods. No observations or outputs are clipped. Bands use the same three cutoffs as the natural-image diagnosis, with DC excluded; gain ties use 10^{-8}, alignment boundaries 10^{-10} and tiny-change energy 10^{-12}. The 240 conditions contain 148 unique inputs and 296 native/zero prediction pairs, not 240 independent images. Inputs and then all predictions were hash-frozen before scoring; this is a deterministic diagnostic, not a blinded natural-image test.

4.2. Alignment reversals and fixed-power counterexamples

For the two middle-frequency templates (32 phase–position conditions), increasing η from 0 to 1 lowers middle-band gain

in 29/32 LTE and 32/32 SC conditions. Harmful outcomes increase from 18 to 32 and 15 to 31, respectively; 14/32 and 16/32 change from positive to negative alignment a. Full64 evaluation retains gain decreases in 31/32 for both models, although exact negative counts change. The high-frequency templates instead contain both signs of path effect (Fig. 1); the path is not uniformly harmful.

At $\eta = 1$, changing phase at fixed template and position produces both positive and negative high-band gains in all eight template–position groups per model, in both evaluation regions. Thus *target power alone does not determine readout benefit in these fixed models*. This does not hold LR power/gradients fixed. At $\eta = 0$, high-phase variants share identical inputs and outputs: score differences then reflect different unobserved targets, not a neural response to hidden phase. These distinctions prevent phase sensitivity from being mistaken for a general adaptive mechanism.

Effects are small: at $\eta = 1$ in the high templates, centre-region full-RGB native-minus-zero PSNR spans $[-.00730, .01236]$ dB for LTE and $[-.00164, .000891]$ dB for SC. Sign reversals provide diagnostic counterexamples, not a practical performance advance or an estimate of natural-image causal prevalence.

5. GUIDANCE, SCOPE AND CONCLUSION

Explanation does not yet supply a selector. For bicubic upsampling $B(y)$, the observable proxy replaces t in Eq. (2); true gain minus proxy gain is exactly $2\langle t - B(y), d\rangle$. Unlike generalized SURE under its stated assumptions [15], this proxy has no unbiased-risk guarantee. With 32 calibration and 12 separate evaluation images (valid0825–0836), forecasts were frozen before scoring. SC's magnitude/spectrum ridge detects 6/14 harmful bands with 15 false positives; adding proxy alignment still detects six with 17 false positives (balanced accuracy $.585$ versus $.568$). Neither candidate establishes a reliable RGB policy gain; LTE has only one harmful band outcome. Saved-output crop/band sensitivity retains SC harm without repairing the forecasts.

A later development test reuses the 32 calibration images, reconstructing each LR observation from a further downsampled version. Eight-fold whole-image-held-out fitting tests whether this lower-level gain transfers. Its within-scale/band macro AUC is $.747$ versus $.601$ for LTE and $.694$ versus $.697$ for SC, but both paired image-bootstrap difference intervals cross zero. Macro averages use the cells with both classes (3/6 LTE, 5/6 SC); bootstrap intervals condition on frozen out-of-fold predictions, without refitting. Choosing native/zero by the lower-level gain increases RGB MSE by $.239\%$ and $.142\%$. This retrospective development does not establish unseen-image generalisation or invalidate all internal-image methods. It was not used to tune the earlier evaluation set.

Scope and cost. The causes of natural-image alignment errors remain unresolved. Synthetic periodicity, a finite phase grid, one checkpoint per model and unsupported zero-cell inputs restrict the intervention evidence. No matching training or new architecture is introduced. Numerical audits verify saved scoring, not an independent network implementation. Saved-output failure analysis and its audit take 2.04 seconds on one CPU thread; controlled inference/scoring takes 85.32 seconds, plus 3.59 seconds preflight and 4.72 seconds audit, on one CPU thread with no GPU. Earlier natural forecasts take 224.19 seconds; internal-transfer development takes 29.68 seconds plus 2.05 seconds audit. These are diagnostic costs, not deployment speedups.

Conclusion. Frequency normalisation improves cross-scale comparison, while pooled curves and coarse band rankings do not determine individual benefit. Harmful cell corrections have distinct overshoot and wrong-direction forms; their importance changes between band and full-image evaluation. Controlled high-frequency paths and fixed-power phase variations expose conditions that frequency energy alone omits. The supported contribution is a reproducible diagnosis of these conditional failures, with global attenuation and the tested predictors retaining explicit counterexamples.

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